EXPERIMENT 14 : Complex number addition using class

A] ALGORITHM;

Step 1: Start

Step 2: Define a class Complex with attributes:

- real
- imag

Step 3: Define method initComplex()

- Read two integers from user input
- Assign them to real and imag

Step 4: Define method sum(c1, c2)

- Set real = c1.real + c2.real
- Set imag = c1.imag + c2.imag

Step 5: Define method display()

- If imag >= 0, print "real + imag i"
- Else, print "real - |imag| i"

Step 6: Create three objects:

- c1, c2, c3 of class Complex

Step 7: Call initComplex() for c1

- Input first complex number

Step 8: Call initComplex() for c2

- Input second complex number

Step 9: Call sum(c1, c2) using c3

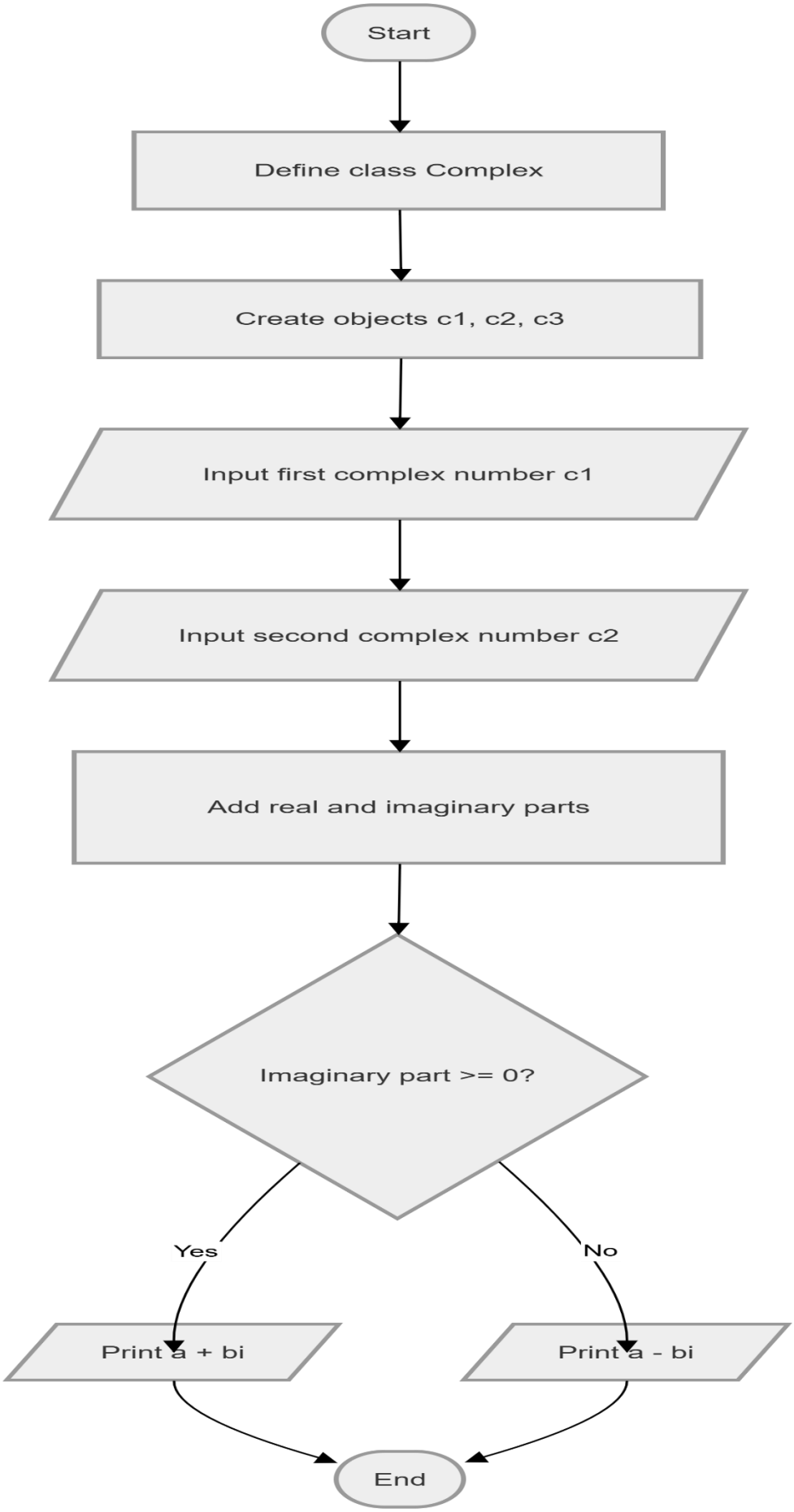
- Store result in c3

Step 10: Call display() using c3

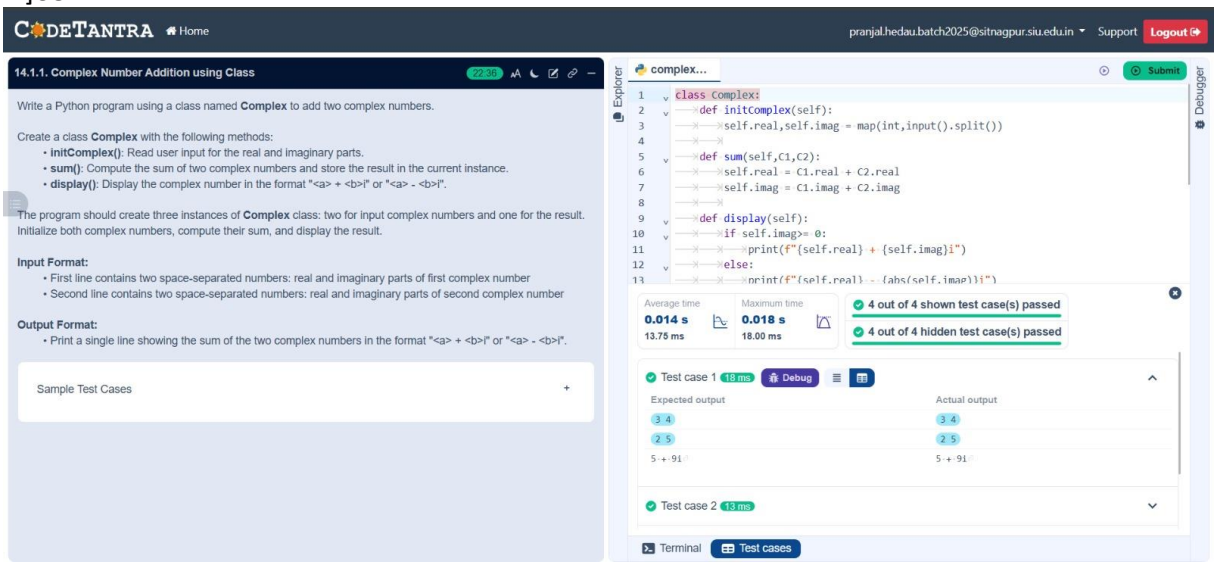
- Display the result

Step 11: Stop

B]FLOWCHART:


Start
Define class Complex
Create objects c1, c2, c3
Input first complex number c1
Input second complex number c2
Add real and imaginary parts
Imaginary part >= 0?
Yes
No
Print a + bi
Print a - bi
End

C] code:

```
class Complex:
    def initComplex(self):
        self.real,self.imag = map(int,input().split())

    def sum(self,C1,C2):
        self.real = C1.real + C2.real
        self.imag = C1.imag + C2.imag

    def display(self):
        if self.imag>= 0:
            print(f"{self.real} + {self.imag}i")
        else:
            print(f"{self.real} - {abs(self.imag)}i")



# Create three instances
c1 = Complex()
c2 = Complex()
c3 = Complex()

# Initialize two complex numbers
c1.initComplex()
c2.initComplex()

# Compute and display sum
c3.sum(c1, c2)
c3.display()
```

D]CODETANTRA: